Kupa Tragy 28 April 2026
BrightLearn
Exercise 01 - SQL FOUNDATIONS

01 Selecting and Sorting.

▸ TABLE employees

Id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

01. Retrieve all columns from the employees table:

SQL Query:

```
SELECT *
FROM employees;
```

Output table:

Id	First_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

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02. Find all unique departments.

SQL query:

```
SELECT DISTINCT department
FROM employees;
```

Output table:

department
IT
HR
Finance
Marketing

03. Retrieve first and last names ordered by salary descending.

SQL query:

```
SELECT First_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

Output table:

First_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

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04. Retrieve the top 3 highest paid employees.
Expected columns: Id First_name last_name Salary

SQL query:

```
SELECT Id,
       First_name,
       last_name,
       Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;
```

Output table:

~~Id la~~

Id	First_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02 Filtering with WHERE

05. Find employees in the IT department.
Expected Columns: Id, First_name, last-name department.

SQL query:

```
SELECT Id,
       First_name,
       last-name,
```

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department
FROM employees
WHERE department = 'IT';

Output table

Id	First_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06, Find employees in Finance with salary > 60000.
Expected columns: Id, First_name, last_name, department, salary.

SQL query:

```
SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance'
  AND salary > 60000;
```

Output table

Id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

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07. Find employees in HR or Marketing.
Expected columns: Id, First_name, last_name, department.

SQL Query:

```
SELECT Id,
       First_name
       last_name
       department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

Output table:

Id	First_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. Find employees not in IT.
Expected columns: Id, First-name, last_name, department

SQL query:

```
~~SELECT Id, First_name,~~
SELECT Id,
       First-name,
       last-name,
       department
FROM employees
```

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WHERE NOT department = 'IT'

Output table

Id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09, Find employees in IT, HR or Finance using IN.

Expected columns: Id first_name last_name, department

SQL query:

```
SELECT Id,
       first_name,
       last_name,
       department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Output table

Id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

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10. Find employees in IT with salary > 65000 and city Johannesburg.

Expected columns: Id, first_name, last_name, department, Salary, city.

SQL query:

```
SELECT Id,
        First_name,
        last_name,
        department,
        salary,
        city
FROM employees
WHERE department = 'IT'
  AND Salary > 65000
  AND city = 'Johannesburg';
```

Output table

Id	First_name	last_name	department	Salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg

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